values of *all* functions of **Q** are conserved. That's what conservation means in quantum mechanics.

The most obvious conserved quantity is the Hamiltonian itself. Since any operator commutes with itself, one can write

$$[\mathbf{H}, \mathbf{H}] = 0$$

which is exactly the condition that **H** is conserved. As in classical mechanics, the Hamiltonian is another word for the energy of a system—it's a definition of energy. We see that under very general conditions, energy is conserved in quantum mechanics.

4.11 SPIN IN A MAGNETIC FIELD

Let's try out the Hamiltonian equations of motion for a single spin. We will first need to specify a Hamiltonian. Where do we get it from? In general, the answer is the same as in classical physics: derive it from experiment, borrow it from some theory that we like, or just pick one and see what it does. But in the case of a single spin, we don't have many options. Let's start with the unit operator I. Since I commutes with all operators, if it were the Hamiltonian, nothing would change with time. Remember, the time-dependence of an observable is given by the commutator of the observable with the Hamiltonian.

The only other choice is a sum of the spin components. In fact, that's exactly what we would get from experimental observation of a real spin—say an electron's spin—in a magnetic field. A magnetic field $\vec{B}$ is a 3-vector—ordinary vector in space—and is specified by three Cartesian components, B_x, B_y, and B_z. When a classical spin (a charged rotor) is put into a magnetic field, it has an energy that depends on its orientation. The energy is proportional to the dot product of the spin and the magnetic field. The quantum version of this is